

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed, developed, and maintained **reliable and scalable data pipelines** and **ETL/ELT architectures** using **Python, SQL** (PostgreSQL, MySQL, Oracle), and **Apache Spark** for distributed data processing; optimized dimensional data models to reduce query latency by **40%** and improved data warehouse performance across financial services operations.
- Architected and deployed **data infrastructure solutions** supporting enterprise business systems and **large-scale distributed data processing**; integrated structured and unstructured data sources across **NoSQL databases** and relational systems, enabling single source of truth for cross-functional analytics and reporting across gTech Ads organization.
- Partnered with cross-functional stakeholders (business analysts, operations teams, data scientists) to translate evolving analytics requirements into actionable technical data solutions; delivered **15+ production-grade reporting environments** and transitioned investigative prototypes into stable dashboards using **Power BI, Tableau**, and advanced **SQL**.
- Authored technical design documents, executed rigorous code reviews, and resolved complex data quality and pipeline bugs; implemented **data validation** frameworks and **schema change management** processes that reduced production incidents by **35%** and ensured regulatory compliance (GDPR, ISO 20022) across banking and financial services infrastructure.
- Developed and maintained **ETLs operating on structured and unstructured sources** using **Python data stack (pandas, NumPy, scikit-learn)** and **Jenkins CI/CD pipelines**; orchestrated end-to-end data workflows including data ingestion, transformation, aggregation, and reporting that processed **100GB+ daily data volumes** with **99.5% uptime SLA**.
- Leveraged **AI technologies to augment and automate the development process** through **Claude API** integration and **prompt engineering** for intelligent data pipeline generation; built AI-driven workflow automation tools that accelerated data solution delivery by **3x** and reduced manual configuration overhead by **60%**.

SKILLS

Data Engineering & Pipelines – ETL/ELT architecture design, data pipeline development, Apache Spark, distributed data processing, data modeling, dimensional modeling, schema optimization, data validation frameworks.

Languages & Querying – Python, SQL (PostgreSQL, MySQL, Oracle), advanced SQL optimization, exploratory data scripts, ad-hoc querying, data warehouse SQL.

Data Infrastructure & Platforms – Data warehouse design, NoSQL databases, large-scale data systems, data infrastructure optimization, single source of truth architecture, database performance tuning.

Data Visualization & BI – Power BI, Tableau, dashboard development, DAX, Power Query, data visualization, business intelligence reporting, metrics definition.

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins CI/CD pipelines, Git version control, code review practices, GitHub Actions.

Data Analytics & Modeling – Statistical analysis, exploratory data analysis, A/B testing, time-series analysis, machine learning fundamentals, scikit-learn, pandas, NumPy, R/RStudio.

Stakeholder Management & Domain – Cross-functional collaboration, requirements translation, project planning and delivery, financial services domain (SWIFT, ISO 20022, SEPA, payments processing, regulatory compliance, GDPR), business systems integration, change management.

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin

02/2026

BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India

01/2021

PROJECTS

SWIFT Payments Operations Data Platform (HCL Technologies)

- Designed and built **scalable data pipeline architecture** for enterprise payments dashboard supporting **SWIFT messaging, ISO 20022 message flows**, and **cross-border payment processing**; ingested and transformed **structured payment transaction data** from multiple sources using **Python ETL scripts** and **SQL** to establish single source of truth for payment operations.
- Developed **dimensional data models** optimized for **synch and asynch system integration** across banking infrastructure; orchestrated **data infrastructure improvements** that reduced reporting latency from **2 hours to 15 minutes** and enabled real-time transaction-flow visibility and operational alerting for SLA management.
- Partnered with banking-domain stakeholders, operations teams, and business analysts to translate requirements into **production-grade reporting environments**; delivered **Power BI dashboards** and **SQL-based analytics** for payments reconciliation workflows, SEPA processing, and regulatory compliance (GDPR, audit trails); ensured data quality through rigorous validation and schema change management across interconnected systems.

Ireland Jobs Tracker – AI-Driven Data Automation Pipeline (Personal Project)

- Architected end-to-end **data pipeline automation** integrating **Python, Claude API** (Anthropic), and **data ingestion workflows**; orchestrated **Apify web scrapers** to extract job description data at scale, transformed unstructured text using **prompt engineering** and structured **JSON ETL processes**, and loaded tailored resume content into **reportlab PDF generation** system.
- Implemented **AI technologies to augment and automate the development process** using **Claude Haiku** and advanced prompt engineering with constraint guardrails; deployed **CI/CD automation** via **GitHub Actions** and **Vercel** to operationalize data workflows, reducing manual intervention by **95%** and enabling on-demand generation of **500+ personalized data artifacts weekly**.
- Designed **data infrastructure** for handling large volumes of unstructured job listing data; applied **exploratory data analysis** and **Python data stack (pandas, NumPy)** to validate output quality and maintain data consistency; demonstrated proficiency in **API integration, workflow orchestration**, and translating business requirements (JD matching) into scalable technical solutions.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)